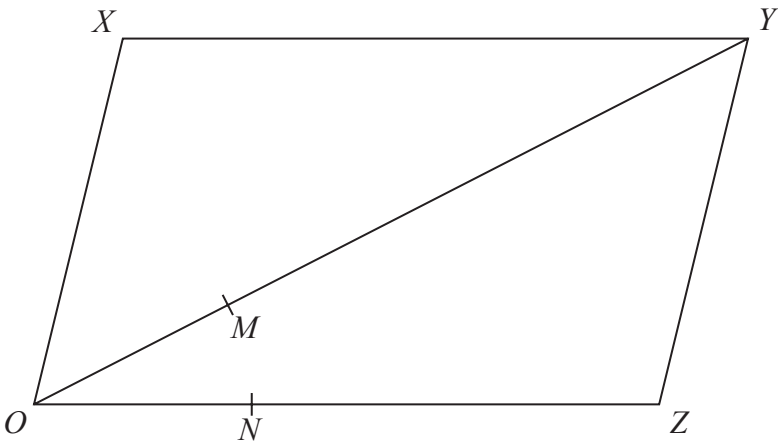


15 $OXYZ$ is a parallelogram.



$\vec{OY} = \mathbf{a}$ and $\vec{OZ} = \mathbf{b}$

M is the point on OY such that $OM:MY = 1:3$
 N is the point on OZ such that $ON:NZ = 1:2$

Work out the ratio $XN:MN$
You must show all your working.